Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam F · Form 1: Pulley Focus

Multiple Choice Questions: Select the best answer. (1 point each)

1. A cart of mass M moves toward a pulley with hanging mass m against constant resistance f. Vary m at fixed total mass S. What feature of a versus m isolates the resistance?

(A) Vertical intercept -f/S

(B) Vertical intercept g

(C) Slope f/S alone

(D) Height g at every point

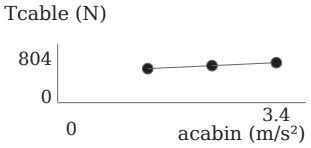
2. Release a 60 kg cabin and adjustable counterweight over an ideal pulley without a motor. Cabin up is positive; g=10 m/s². Halve cabin mass and produce the same cabin acceleration values. Plot the difference Ynew-Yold versus X. What slope should be observed?

(A) -17.4

(B) -55.2

(C) 7.8

(D) -30



3. For an Atwood machine, students measure a but neither mass. Without a second measurement what can they infer, assuming an ideal device and known g?

(A) The tension uniquely

(B) Only the sum of the masses

(C) The mass ratio but not the individual mass scale

(D) Both masses uniquely

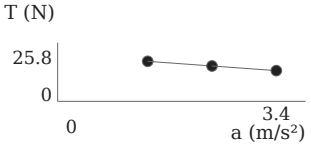
4. An unseen hanger rides on one side of an ideal pulley. Vary the opposite load and record tension with the unseen hanger acceleration positive downward. Use g=10 m/s². Replace the unseen hanger by twice its mass at the same accelerations. Which X in the modified apparatus yields Y=45.6?

(A) 1.14

(B) 4.29

(C) 2.4

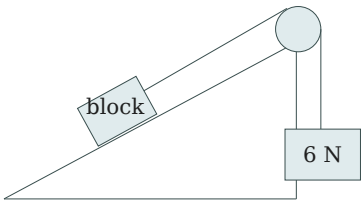
(D) 3.03



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A 10 N block rests on a 30° incline with a 6 N hanger. Apply vertical downward force P to the block. Static friction suffices.

a. Separate hanger and block; distinguish the applied P from gravity. **(1 pt)**



b. Find the reversal point and derive the minimum static coefficient as a function of P. **(3 pt)**

6. An incline block of weight 10 N rests on a 30° ramp with a 5 N hanger on an ideal pulley. Add vertical downward force P to the block. Friction is sufficient; uphill is positive. P (N) -> fs signed (N): (1.4, 0.7); (2.4, 1.2); (3.4, 1.7).

a. Build the model for pressing on an incline from physics and predict both slope and intercept. **(2 pt)**

b. Change the tilt so sinθ increases by a factor of 1.5. Static friction remains sufficient. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. A stationary incline block has mass M and ramp angle θ. A vertically downward extra force P acts on the block; an ideal rope connects it to hanger m. Derive the minimum static coefficient for rest without assuming a friction direction. **(3 pt)**